Similarity Judgments Reveal the Multidimensional Representations of Naturalistic Faces

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Abstract

The human face affords a remarkable array of complex social cues, yet our understanding of how it is spontaneously perceived relies almost exclusively on controlled, posed stimuli that suppress the natural variation which makes faces so informative. Here, using a data-driven similarity task with 75 naturalistic face images, we mapped 182,500 perceptual judgments across 365 adults aged 20 to 76. The results indicated that naturalistic face space was hierarchically organized by expressive and demographic cues: smiling faces formed a coherent category that transcended gender and age, whereas non-smiling adult faces stratified along apparent gender. Representational similarity analysis showed that affective dimensions (valence, intensity, energy), emotion category, demographic cues (age and gender), and image-based color each uniquely predicted similarity judgments, while specific personality traits and facial action unit intensity did not. These results demonstrate that highly variable naturalistic faces carry potent combinations of both categorical and dimensional information, advancing a more ecologically valid framework for understanding real-world social perception.

Keywords: face perception; social cognition; emotions; social categorization; facial expressions

Statement of Relevance

Faces provide an abundance of cues essential for successful social coordination. Yet much of our understanding of face perception comes from experiments using posed faces and tasks that require observers to judge predefined categories. Consequently, we know relatively little about how people spontaneously organise the complex socially relevant information conveyed by naturalistic faces. Here, we asked adults to judge which faces appeared most similar, without specifying which attributes should guide their decisions. Despite considerable variation in the images, observers made remarkably consistent judgments that reflected combinations of emotional, demographic, and visual information. These findings reveal that naturalistic face perception is both multidimensional and systematically organised, demonstrating how data-driven approaches can uncover the information people spontaneously prioritise when interpreting faces.

Highlights:

- Naturalistic faces elicit highly consistent similarity judgments across adults
- Valence, emotion category, age and gender uniquely predict face similarity
- Smiles form a coherent category that cuts across gender and age
- Variable naturalistic faces convey both categorical and dimensional information

Research Transparency Statement

General Disclosures

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Preregistration:

The hypotheses, experimental methods, and analysis were preregistered (https://aspredicted.org/4P8_C88) prior to data collection. There were minor deviations from the preregistration (for details see Supplementary Table 1). All study materials, data, and analysis code are publicly available through Anonymous Github (https://anonymous.4open.science/r/multidimensional-DEF8).

Authorship:

N.P. played a lead role in project administration, data curation, formal analysis, investigation, visualization, writing–original draft, writing–review and editing, validation, an equal role in methodology, software and resources. A.R. played an equal role in software and resources, a supporting role in writing–review and editing and methodology. T.G. played a supporting role in methodology, software, and resources, and writing – review and editing. J.D.H. played a supporting role in conceptualisation and writing – review and editing. J.T. played a lead role in conceptualisation, funding acquisition, an equal role in methodology, resources, and a supporting role in writing – review and editing.

Multidimensional Representations of Naturalistic Faces

Making sense of the faces we see is an extraordinary feat. From a deceptively simple configuration of facial attributes, even the faces of strangers convey vast quantities of information, such as age, gender, identity, physiological state, personality traits, emotional appeals, and conversational intent (Buckland & Taubert, 2025; Cowen & Keltner, 2020; Holler, 2025; Leung et al., 2023; Liu et al., 2022; Scarantino et al., 2022; Sutherland & Young, 2022; White & Burton, 2022). To extract this multidimensional information, the brain is thought to compare incoming facial attributes against stored mental representations (Valentine et al., 2016). To isolate specific mechanisms, previous research has traditionally relied on tightly controlled face stimuli, varying attributes along isolated dimensions, such as identity, expression, gaze or face shape, while holding all other facial attributes constant. While such studies have provided foundational insights into how faces are categorized (Bruce et al., 1987; Bruce & Young, 1986; Porcheron et al., 2013; Rossion, 2002; Taubert et al., 2016), recognized (Taubert & Alais, 2011; Towler et al., 2023; White & Burton, 2022), elicit social inferences (Barrett et al., 2019; Fujimura et al., 2012; Russell, 1980), and guide predictions (Chanes et al., 2018), they necessarily restrict sampling to a curated subset of representational space. Consequently, these paradigms may not fully account for the variable and flexible feature integration characteristic of facial information 'in the wild'.

Further, traditional methods for eliciting behavioral judgments typically assess performance using categorization or recognition tasks, which make it difficult to elicit behavioral judgments without biasing responses toward predefined dimensions. For example, facial emotion recognition paradigms often rely on prototypical emotion categories (happiness, anger, fear, disgust, sadness, and surprise) represented by distinct configurations of facial action units (AUs) (Ekman & Friesen, 1978). Stimuli are typically controlled for these activations and paired with corresponding labels to assess behavior in laboratory settings. While highly valuable for establishing baseline mechanisms, this approach necessarily leaves aside non-prototypical AU configurations and constrains the diversity of descriptive

labels available to participants. Restricting response options in this way has been shown to influence performance, by potentially inflating consensus (Adams & Weisbuch, 2026; Nelson & Russell, 2013), limiting observations to predefined categories (Betz et al., 2019; Lindquist & Gendron, 2013), and/or privileging singular dimensions over latent, complex, or contradictory social cues (Matsumoto & Wilson, 2026). Because natural faces reflect multiple interacting dimensions, constrained approaches risk inadvertently reinforcing the theoretical commitments they intend to test (Peluso et al., 2025). Consequently, less is known about how mental representations of naturalistic faces are organized when facial signals retain information traditionally treated as noise.

More recently, data-driven methods have emerged for assessing perceptual face similarity in an unbiased way, making it possible to use larger and more complex stimulus sets which seek to explain face perception under more ecologically valid conditions. Here, our goal was to use one such method to characterize the multidimensional representations underlying a set of “uncontrolled” faces without imposing a facial attribute of interest. To this end, we quantified perceptual similarity using a triplet odd-one-out task previously used to understand latent dimensions in objects (Grootswagers et al., 2024; Hebart et al., 2020) with the Wild Faces Database (Peluso, Long et al., 2023), a diverse collection of ambient images displaying a range of heterogeneous facial gestures. The triplet odd-one-out task presents three stimuli simultaneously, with the participant choosing the one that is most different. No further instructions are provided to avoid biasing attention to any singular dimension, allowing the task to reveal similarities which might otherwise remain latent without presupposing or labeling the attributes of interest in advance.

In order to characterize the multidimensional space revealed by this paradigm, we separately collected behavioral ratings for 75 Wild Face images, consisting of 70 human faces (35 male, 31 female, four ambiguous) and five illusory faces (pareidolia), which we retained given evidence that perceiving face-ness in objects influences the perception of social cues such as valence and gender (Saurels et al.,

2024; Taubert et al., 2023; Wardle et al., 2022). These 75 faces were independently rated using traditional emotion category labels (happiness, sadness, anger, fear, disgust, surprise, neutral), and along twelve other dimensions previously identified as important for social cognition (Adolphs et al., 2016; Baudouin et al., 2025; Bruce et al., 1987; Lin et al., 2025; Peluso, Long et al., 2023; Norton & Pettegrew, 1979; Porcheron et al., 2013; Todorov et al., 2011). We included categories perceived from relatively stable facial attributes (chronological age, gender, gaze direction), and dimensions derived from more variable facial features such as affective inferences (valence, intensity, energy), and social traits (attentiveness, attractiveness, familiarity, genuineness, likeability, and trustworthiness).

We then used Representational Similarity Analysis (RSA), Multidimensional Scaling (MDS), and hierarchical clustering to map this representational space. These allowed us to directly compare participants' similarity judgments against our socio-evaluative ratings, low-level image properties using Gabor-Jets, VGG Face, CIE L*a*b* color and luminance (Grootswagers et al., 2019; Parkhi et al., 2015; Yue et al., 2012), and computationally derived descriptions of facial activity (OpenFace 2.0; Baltrusaitis et al., 2018)). Crucially, we identify a remarkably robust geometry underlying naturalistic face similarity judgements, scaffolded by interpretable and flexible combinations of affective dimensions and social categories. The results provide an important and necessary step toward an ecologically informed model for describing how humans make sense of faces in the real world.

Materials and Methods

Participants

We recruited 365 healthy participants between 20 and 76 years of age (182 females, 1 non-binary; *M = 41.95, SD* = 13.64) via Prolific (www.prolific.com) to participate in an online experiment. To ensure we had enough power to examine whether representations would vary as a function of age, recruitment was streamed by nine age groups (20-24, 25-29, 30-34, 35-39 etc.) to ensure each group sampled a minimum of 40 participants containing an equitable distribution of gender. This number was

derived through pilot testing to ensure there were at least 20 observations for each pairwise similarity comparison (see Supplementary Text 1). Participants were excluded if they were previously diagnosed with neurological conditions. Given the risk of non-human agents participating in online experiments, we manually screened participants for implausibly fast task completion (< 20 minutes), and none were excluded. Participants were primarily based in the UK, Canada, and the United States and were paid approximately $21 AUS (£11 GBP) for their participation, for which they provided informed consent. The study was approved by The University of Queensland's Human Research Ethics Committee (2024/HE000198).

Stimuli

Stimuli comprised 75 images (hereafter referred to as the WFD-75) selected from the Wild Faces Database using a computational Variational Interpretable Concept Embeddings (VICE) model (Muttenthaler et al., 2022). This model previously identified 15 dimensions underlying triplet odd-one-out judgments across 900 WFD images, excluding animal faces (Taubert et al., 2024). To ensure broad coverage of facial attribute space, the top five exemplars from each dimension were selected comprising 70 human faces (35 male, 31 female, four ambiguous) and five objects with face-like qualities (pareidolia). Images were 400 x 400 pixels and varied widely in head orientation, gaze direction, color profile, and image quality.

Procedure

Triplet Odd-One-Out Task

Visual similarity judgments were acquired remotely using a triplet odd-one-out task previously used to examine representational geometry in natural objects and human actions (Bockes et al., 2025; Grootswagers et al., 2024; Hebart et al., 2020). The task was programmed in jsPsych (de Leeuw, 2015) and hosted on Pavlovia (pavlovia.org). Each trial displayed a random combination of three faces arranged equidistant from each other from the center of the screen (Fig. 1). Participants were asked to

select one image as the odd-one-out, as being the image they judged to be most different to the other two. No additional instructions were provided to avoid biasing to particular features, and each triplet remained visible until a choice was made by responding with a mouse click on the chosen image, initiating a 500 ms interval before the next trial. Breaks were provided every 201 trials, with participants completing 500 trials yielding a total of 182,500 similarity judgments across the entire experiment.

Socio-Evaluative Ratings Task

To characterize visual information in the WFD-75, independent ratings were collected from a separate sample of 75 undergraduates (age 17-25; 47 females, 27 males, 1 undisclosed) recruited via The University of Queensland research participation scheme, who volunteered in exchange for course credit. Participants were asked to rate each image along dimensions of attentiveness, attractiveness, energy, familiarity, likeability, and trustworthiness using a continuous Likert sliding scale from 1 on the left to 9 on the right (for example, “How likeable/attractive/trustworthy is this face?”; 1 = Not at all, 9 = Extremely), before advancing to the next image. Gaze ratings were obtained with the question “Which direction is the face gazing? (Report the direction from your perspective. If the eyes are closed, please guess)” with 1 = To the left, 5 = direct, 9 = To the right. Age ratings were obtained using a continuous slider from 0 – 100. Between 25 and 42 judgments were collected for each rating and the study was approved by The University of Queensland’s Human Research Ethics Committee (2024/HE000198). Additional ratings for gender, valence, emotion category, intensity, and genuineness were obtained from previous studies described elsewhere (Peluso, Long et al., 2023; Wei et al., 2026, see Supplementary Text 2).

Analysis

Representational Similarity Analysis (RSA)

Behavioral data were analyzed in MATLAB 2025b using the Representational Similarity Analysis (RSA) framework, a multivariate analysis used in previous tasks to characterize patterns of responses to

visual stimuli (Kriegeskorte et al., 2008). Odd-one-out judgments were used to calculate dissimilarity between each pair of experimental stimuli (Fig. 1). For each pair of faces, similarity was computed as the proportion of trials where the pair appeared together and neither image was selected as the odd one out, divided by the total number of trials that included that pair. These similarity scores were then subtracted from 1 for conversion into dissimilarity scores, such that higher values indicated pairs were perceived as more different. Dissimilarity scores were calculated for all pairs of 75 stimuli, resulting in a 75 x 75 Representational Dissimilarity Matrix (RDM) comprising 2,775 unique pairs.

We also used RSA to compute the split-half consensus of similarity judgments across participants by randomly dividing participants into two groups and correlating their RDMs across 1,000 bootstrap iterations. Using the same bootstrap method, we also assessed between-trial reliability by dividing trials into odd- and even-indexed halves, retaining the full *N* in each half. Correlations were corrected using the Spearman-Brown formula.

Model Representational Dissimilarity Matrices. We built multiple model RDMs for perceived age, attentiveness, attractiveness, energy, familiarity, gaze, gender, genuineness, intensity, likeability, trustworthiness and valence by computing pairwise dissimilarity as the absolute difference between the two faces' mean ratings (using city-block distance, which for a single rating dimension is identical to Euclidean distance). The emotion category RDM was constructed using dissimilarity in the rated emotional profile of each face. First, for all 75 faces we constructed a 9 x 1 vector based on the proportion of raters who selected each emotion category label (anger, fear, disgust, sadness, neutral, happiness, surprise, other or unknown) for each face in a previous labeling task (Peluso, Long et al., 2023). Dissimilarity between each pair of faces was 1 − Pearson's r between their vectors.

Computational image-derived model RDMs were built using numerical vectors that described image luminance and color distribution (both using CIE L*a*b* color space), other image features extracted from Gabor-Jets (Lades et al., 1993) and VGG Face (Parkhi et al., 2015), and facial action unit

intensity detection using OpenFace 2.0 (Baltrusaitis et al., 2018). These models allowed assessment of the low-level and facial movement features previously found to correlate with visual similarity. For exact model specifications see Supplementary Text 2.

Statistical Inference Using Multiple Regression on Distance Matrices

We used multiple regression on RDM distance matrices to estimate each candidate model's unique contribution to behavioral similarity with all other models held constant. The behavioral RDM for all participants was vectorized and regressed simultaneously onto the corresponding vectors from each of our candidate models as predictors, yielding standardized regression coefficients (β) reflecting each model's unique contribution after partialling out shared variance. One challenge with inference using RSA is that observations within an RDM are not independent. We therefore followed a stimulus-label randomisation approach (Kriegeskorte et al., 2008; Nili et al., 2014) and combined this exchangeability method with the Freedman-Lane procedure (Winkler et al., 2014), under which each predictor is tested against a null that retains the effects of the remaining predictors. Permutation *p*-values were corrected for multiple comparisons using the Benjamini–Hochberg false discovery rate. Analysis was conducted using RStudio 2026.01.1+403.

Multidimensional Scaling (MDS)

We used Multidimensional Scaling (MDS) to map the underlying structure of the behavioral RDM, such that faces perceived as more similar are positioned closer to each other in space than faces perceived as more different (Kruskal & Wish, 1978). To assess the optimal number of dimensions, we computed MDS solutions from 2 – 8 dimensions using Kruskal's Stress 1 formula to measure fit quality, achieving stress 0.077 by seven dimensions but with no clear elbow (see Supplementary Fig. 1). Given improvement dropped to ~1.2% between seven and eight dimensions, we retained the seven-dimensional (7D) solution as an optimal balance between preserving meaningful structure while retaining interpretability.

Hierarchical Clustering Analysis

We applied hierarchical clustering (average linkage) using the pairwise Euclidean distances derived from the 7D MDS embedding using MATLAB 2025b. Average linkage defines the similarity between two clusters as the average distance between all possible pairs of observations across the two groups. Given naturalistic face stimuli are more perceptually graded, this algorithm was chosen for its robustness to outliers and to preserve the geometric structure of the underlying pairwise distances. We evaluated the optimal number of clusters by computing mean silhouette width for solutions from k = 2 to 10, using Euclidean distances in the same seven-dimensional space in which the clustering was performed. Silhouette width was maximal for the trivial two-cluster solution ($\bar{s}$ = .35), which separated the five pareidolia stimuli from the remaining faces, while in the range of human faces silhouette width rose from a minimum at k = 3 ($\bar{s}$ = .20) to a maximum at k = 7 ($\bar{s}$ = .29), with k = 8 closely comparable ($\bar{s}$ = .28) before plateauing to k = 10 ($\bar{s}$ = .28). Given the graded nature of the silhouette values, we also examined the merge heights of the dendrogram itself, where two face images high in similarity are located in close proximity and connected by vertical branches which are low in height. Excluding the human/pareidolia cluster split (gap = .185), the largest increases occurred at k = 4 (gap = .055) and k = 8 (gap = .044); no other solution exceeded .025. We therefore converged upon an eight-cluster solution which explained 66.59% of the total variance in the 7D MDS space, representing a robust clustering solution with strong overall separation. Clusters were profiled using the independent ratings data and OpenFace 2.0 using RStudio 2026.01.1+403.

Results

Naturalistic faces elicit robust consensus in similarity judgments across the adult lifespan

We quantified consensus among participants using split-half reliability and found similarity judgments were remarkably consistent both across participants (r = .968) and across trials (r = .969), Spearman-Brown corrected. Given the equitable distribution of female and male participants within the

full sample (n = 182 each), we also quantified consensus within and between each gender group which we found to be similarly high (females: across-participants, r = .938 and across-trials, r = .946; males: across-participants, r = .936 and across-trials, r = .939). Similarly, the behavioral RDMs that were generated by male and female participants were highly correlated with each other (Spearman's rho r = .92, $p < .001$).

To investigate any potential age differences in similarity judgments, we also verified consensus by grouping data by participant age (20-24, 25-29, 30-34, 35-39, 40-44, 45-49, 50-54, 55-59, 60-64+) to construct nine separate behavioral RDMs. Each age RDM contained 40-43 participants, each covering 100% of 2,775 pairwise comparisons with a median count of 21-23 observations. Despite the relative sparsity in these RDMs compared to the full sample RDM (which contained a median count of 149-197 observations per pair), consensus remained strong with all age groups significantly correlated with each other, ranging between r_s = .697 and .764, $p < .001$, FDR corrected.

Independent predictors of naturalistic face similarity

After establishing impressive agreement between participants as to which faces were more similar than others, we next sought to identify which of our socio-evaluative ratings and stimulus-derived computational models might independently predict these judgments using multiple regression. Given our task included illusory faces (i.e., pareidolia) which do not feature facial action units and may contain contrasting visual statistics to human faces (such as luminance and color), we conducted three separate analyses: the first included all 75 stimuli (70 human faces and 5 pareidolia) with 17 predictors, excluding OpenFace. The second introduced OpenFace and excluded six stimuli OpenFace could not process (five pareidolia stimuli and one black-and-white human face), yielding a 69-face dataset with the full 18-predictor set. Given Analyses One and Two differed in both stimulus set and predictor set, Analysis Three was fitted on the 69-face dataset with the same predictors as Analysis One to be able to compare the effect of stimulus removal separately to the inclusion of OpenFace. Overall model fit was

significant in all three analyses (permutation $p < .001$; $R^2 = .351$, .371, and .369, respectively), corresponding to approximately 36–39% of the explainable variance given the noise ceiling ($R^2 = .969$ and .957 for 75 and 69 stimuli).

In Analysis One (75 stimuli, all predictors except OpenFace), eight predictors contributed independently to behavioral similarity after FDR correction: genuineness (β = .268), emotion category (β = .204), age (β = .158), valence (β = .156), color (β = .154), gender (β = .143), luminance (β = .100), and energy (β = .072; Fig. 2, orange lines). In Analysis Two (69 human faces, including OpenFace), the significant predictors were valence (β = .254), emotion category (β = .238), gender (β = .197), age (β = .163), intensity (β = .134), luminance (β = .120), energy (β = .066), and color (β = .064), whereas OpenFace was not significant (β = .052; Fig. 2, green lines). Analysis Three (69 human faces, all predictors except OpenFace) yielded the same eight significant predictors with comparable coefficients (Fig. 2, blue lines), and OpenFace's unique contribution was negligible ($\Delta R^2 = .002$). The differences between Analysis One and the 69-face analyses therefore reflect the removal of pareidolia rather than the inclusion of OpenFace: genuineness no longer contributed (β = −.026), color was attenuated (β = .064), and intensity emerged (β = .135). This suggests that the genuineness and color effects in Analysis One may partly reflect the perceptual distinction between human and illusory faces. Together, these results indicate that naturalistic face similarity is predicted by a combination of expressive (valence, emotion category, intensity, energy), demographic (age, gender), and image-level (luminance, color) dimensions. By contrast, in the 69-face analyses, none of the remaining socio-evaluative ratings (trustworthiness, likeability, attractiveness, attentiveness, familiarity, gaze, genuineness) or computational models (OpenFace, Gabor-Jet, VGG) explained unique variance after FDR correction (see Supplementary Tables 2–4).

Hierarchical clustering reveals interpretable organization of naturalistic face space

Our results so far reveal naturalistic faces elicit high-dimensional and remarkably consistent similarity judgments which are uniquely predicted by expressive, demographic, and image-level cues. To identify the face attributes underlying these dimensions, we used hierarchical clustering as a means of recovering categorical structure while preserving dimensionality, grouping stimuli solely on the basis of their pairwise dissimilarities. Figure 3 displays the full clustering solution as a vertical dendrogram alongside the RDM arranged in the order of images derived from hierarchical clustering, revealing highly distinct clusters of low dissimilarity along the diagonal (dark blue). Pareidolia images (*n* = 5) emerged as the most distinct cluster and were judged as strongly dissimilar to human faces, visible as a high dissimilarity (yellow) horizontal stripe along the bottom of the RDM.

Accordingly, if the resulting clusters are interpretable, then face images should share certain identifiable properties that might explain category membership. In Figure 4, we inspected the face images within each cluster alongside the dendrogram solution, and found each cluster contained interpretable, yet complex combinations of visually similar stable and dynamic social cues. We describe these groups in terms of their shared features.

The most distinct cluster (*n* = 5, blue) contained all face pareidolia images (Fig. 4, top), indicating a strong categorical boundary between human faces and illusory faces in objects which emerged at the highest branching point of the dendrogram (~0.8). Among human faces, the most distinct cluster contained images of children/babies (*n* = 7, cyan) irrespective of variability on other dimensions (e.g. gender, facial gesture). Indeed, mean attributed age for this group of stimuli was 5.67 (*SD* = 1.84) in comparison to the mean age of the remaining stimuli (which ranged from 21.86 - 39.14). Interestingly, older faces did not similarly form a unique cluster, but were distributed into other adult clusters, suggesting an asymmetry in how age information is combined with other social cues. Cluster three revealed a uniquely coherent category for smiles (*n* = 12, green) which transcended variations in stable demographic cues such as gender, ethnicity, and age, and highly variable cues such as gaze direction and

head orientation. Remaining clusters were organized by salient non-smiling facial gestures (*n* = 13, teal), faces with hands covering mouths (*n* = 3, orange), or by gender and/or ethnicity (*n* = 11, red; *n* = 15, vermillion; *n* = 9, purple). Interestingly, while the perceived gender and ethnicity of face stimuli seemed robust organizers of clusters six to eight, these cues also seemed important within clusters two to five, with faces from the same gender or ethnicity tending to be proximal to each other.

Demographic and Affective Dimensions Form Distinct Spatial Gradients

To understand how dimensions were distributed across the global embedding as well as within each cluster, we first mapped the coordinates derived from the seven-dimensional solution onto two-dimensional multidimensional scaling (MDS) solution and colored each stimulus in the embedding according to its cluster membership (Fig. 5). We then repeated this process with the face stimuli themselves. This revealed that each cluster was spatially coherent and well differentiated even in this low dimensional simplification.

We then used the same method to examine how individual dimensions were spatially organized across the same embedding by recoloring each stimulus according to our ratings of perceived age, gender, and valence (Fig. 6). Here, each face image is represented as a colored dot on the embedding according to its unstandardized value on each dimension and presented alongside the same values visualized as boxplots according to cluster. By doing so, we found that cluster specificity in each dimension mapped to distinct spatial gradients across the global embedding. First, while age evaluations were noticeably younger in cluster two (*M*age = 5.67, where age was rated from 0 – 100) these images did not cluster together when mapped in two-dimensional MDS space, but rather were distributed along a vertical gradient (Fig. 6a). Intriguingly, older faces were not located directly opposite the youngest faces on the embedding, but rather adjacent in a similar vertical distribution. Similarly, mean gender ratings per cluster (Fig.6b) confirmed the highest gender specificity in clusters six (*M*gender = 2.53, where 1 = extremely feminine) and cluster seven (*M*gender = 7.56, where 9 = extremely masculine), yet

the embedding revealed a distinctly different spatial organization with feminine faces vertically concentrated in the center, masculine faces to the left and androgynous faces to the right (Fig. 6b). Valence ratings (Fig. 6c) also revealed strong cluster-specific alignment, particularly in cluster three (smiles, *M*valence = 7.2) where highly positive faces formed a dense, well-defined region in MDS space. At the opposite extreme to cluster three, cluster four contained face images rated as more negative (*M*valence = 3.27) confirming the presence of a graded valence axis through MDS space, with the center however remaining more diffuse.

Using the same method, we further discovered that arousal-related dimensions of energy and intensity were not independent of valence (Fig. 7). Both scatterplots and recolored MDS plots showed a similar V-shape pattern across the embedding for both energy and intensity dimensions, with lowest values in face pareidolia and rising toward both the high-valence region (smiling faces, cluster three) and the low-valence region (cluster four). See Supplementary Table 5 for all socio-evaluative ratings by cluster, Supplementary Fig. 2-4 for boxplots and recolored MDS plots for all socio-evaluative ratings.

High Heterogeneity in Emotion Category Labels and Facial Action Units Within Clusters

Finally, given that specific facial AU configurations are thought necessary for emotion categorization, this seemed at odds with our finding that facial AU intensity (as detected by OpenFace) did not significantly predict naturalistic face similarity. Accordingly, it was possible that our clustering might reveal the source of this apparent dissonance. To investigate, we first visualized the percentage of face images for each cluster that corresponded to each of the same nine emotion categories that were rated by our independent sample used to build the emotion category model (anger, disgust, fear, happiness, sadness, surprise, neutral, other, or unknown; Fig. 8a). This revealed that clusters were not associated with coherent emotion category labels, with the remarkable exception of cluster three (smiles) which showed near-exclusive attributions for happiness. This specificity was also clear when emotion category labels were mapped to MDS coordinates (Fig. 8b).

We then asked whether this emotion category heterogeneity extended to facial action unit (AU) detection profiles by calculating the coefficient of variation (CV = *SD/M*) from the 69-stimuli OpenFace AU intensity detections across cluster members (Fig. 9) as well as the proportion of cluster members with active AUs and the mean intensity for each AU by cluster (Supplementary Fig. 5 - 6). Again, we found evidence of considerable variability within clusters, characterized by predominately high CV values (1.0–2.8). While cluster three (smiles, *n* = 12) was notably coherent across both AU12 (Lip corner puller, CV = 0.5) and AU06 (Cheek raiser, CV = 0.5) – activations thought crucial for so-called genuine or Duchenne smiles - cluster five (*n* = 3) also revealed similar specificity across several mouth-related AUs including AU14 (Dimpler, CV = 0.5), AU07 (Lid tightener, CV = 0.6), and AU20 (Lip stretcher, CV = 0.3) despite these images all containing faces with hands covering the mouth region. This confusability suggests challenges remain in detecting AUs in naturalistic face stimuli, and that naturalistic face stimuli may contain more complex combinations of AUs than basic emotion category models predict.

Discussion

In this study we combined an unbiased measure of perceived similarity, data-driven multivariate analyses, and independent perceptual and socio-evaluative explanatory models, to map the similarity across 75 naturalistic face images, allowing the stimuli to freely vary without specifying in advance which dimensions should matter. Three major findings emerged. First, we found impressive agreement in how observers spontaneously responded to highly variable naturalistic facial attributes across the adult lifespan, even without providing explicit instructions. Second, we found that models evaluating the expressive (valence, emotion category, intensity, energy), demographic (facial age, gender), and image-level (luminance, color) dimensions of naturalistic faces partially captured this representational geometry. Third, we found these dimensions were flexibly combined to structure the underlying multidimensionality of naturalistic face space; cues consistent with facial age and gender provided pronounced categorical structure, yet smiles were elevated to cut across these stable demographic

cues. Together, these findings reveal a hierarchically organized yet context-sensitive geometry for incorporating multiple behaviorally relevant social cues in naturalistic faces, with important consequences for understanding the foundations underlying human face perception.

That observers spontaneously extracted multiplexed social categories and dimensions from uncontrolled naturalistic faces is consistent with a contextually sensitive process for integrating social affordances from stable and expressive facial attributes using more controlled stimuli (Baudouin et al., 2025; Fujimura et al., 2012; Liu et al., 2022; Mehu and Scherer 2015). Here, we extend this work to reveal that categories such as gender, age, and emotion were not uniformly defined across naturalistic face space but rather interacted differently depending on the presence or absence of salient expressive cues. Further, dimensions of intensity and energy were not orthogonal to bipolar valence but instead appeared dependent, consistent with a V-shaped relationship between valence and arousal-relevant information (Kuppens et al., 2013). Together, these findings challenge the assumption that increasing variability in facial appearance necessarily compromises the separability or reliability of categorical and dimensional information (Dawel et al., 2022; Peluso et al., 2025; Ritchie & Burton, 2017). Instead, we demonstrate that even when observers are given no instructions about which facial attributes to evaluate, naturalistic faces retain a multiplexed, yet highly consistent and interpretable perceptual structure (Sutherland et al., 2013). It follows that difficulties in interpreting naturalistic expressions may arise not simply from variability in the stimuli themselves, but from experimental approaches that require observers to reduce complex, multidimensional facial information to discrete, predefined categories (Barrett et al., 2019; Hoemann & Feldman Barrett, 2019; McVeigh et al., 2026).

Importantly, we also found that this multiplexing was not associated with similar coherence in facial AU activations, consistent with evidence that the same facial AUs facilitate multiple dimensional and categorical inferences (Mehu and Scherer, 2015; Liu et al., 2022). Indeed, we found the activations themselves were considerably heterogeneous, which may reflect measurement noise. For example, we

found confident detections of mouth-specific AUs in faces with no visible mouth (e.g., hand-over-face stimuli), and noticeably lower detections for faces shown in profile, highlighting potential challenges for computational AU detection in generalising across occlusions and other sources of variation in ambient pictures of faces (Cross et al., 2023). Alternatively, this heterogeneity may reflect genuine differences in facial appearance that are captured by OpenFace but are less relevant to human judgments of perceptual similarity. This possibility raises an important distinction between the physical features detected by computational models and the information prioritised by human observers. Indeed, AU activations varied considerably even among faces judged to be highly similar (Supplementary Fig. X). One exception was smiling faces, which exhibited relatively consistent activation of AU06 (cheek raiser) and AU12 (lip corner puller), suggesting that OpenFace can recover coherent facial configurations despite naturalistic variability. Moreover, the emergence of smiling faces as a distinct perceptual cluster, transcending differences in age and gender, suggests that smiles provide a particularly salient and consistent cue that observers spontaneously prioritise amid competing facial information (Peluso et al., 2025), rather than an artifact of experimentally constrained stimulus categories. In sum, these findings suggest that the relationship between physical facial configurations and perceptual similarity is not uniform with some expressions, such as smiles, exhibiting coherence at both levels, whereas other perceptually similar faces may be supported by more heterogeneous combinations of facial features.

Our approach is not without limitations; we make no claim that our stimulus set represents the full spectrum of naturalistic facial attributes, and a different set of ambient faces could produce a distinct representational topography. However, we intentionally chose a compact, highly diverse stimulus set in contrast to data-driven approaches operating at much larger, sparser scales (Bockes et al., 2025; Hebart et al., 2020). This design facilitates future fMRI and E/MEG investigation, enabling dense repeated sampling of the same images across different populations to isolate contextual effects and individual differences that remain understudied (Hehman et al., 2017; White & Burton, 2022).

Accordingly, future work should prioritize testing non-Western cohorts and scaling this paradigm to larger stimulus sets to determine how broadly this representational geometry generalizes.

While traditional approaches to face perception have often sought to minimise stimulus variability, our findings demonstrate that naturalistic faces can elicit remarkably consistent and reliable perceptual judgments, even across observers aged 20 to 76 years of age. These findings suggest that the challenge for understanding face perception is not simply to control variability, but to identify the perceptual regularities that emerge despite it. This distinction has important implications for both computational models and applied measures of social cognition. For example, incorporating the multidimensional structure of naturalistic face representations into computational models may help develop more robust and generalisable facial analysis tools, with potential implications for reducing bias in automated decision-making. Similarly, conventional clinical assessments of social cognition often rely on posed facial expressions and forced-choice categorisation tasks, which may not adequately capture how individuals interpret complex or ambiguous facial information in everyday life. Establishing perceptual benchmarks grounded in naturalistic face similarity judgments may therefore provide a more ecologically valid framework for evaluating individual differences in social cognition (Henry et al., 2023).

Taken together, we demonstrate that humans spontaneously and flexibly combine broad categories and fine-grained dimensions to evaluate naturalistic faces. By advancing a new methodological framework that accommodates variation in both facial information and observer behavior, this work moves beyond traditional stimulus rewilding (Dawel et al., 2025; Peluso, Long et al., 2023; Peluso et al., 2025) toward data-driven paradigms that allow the complex, flexible structure of face representations to guide more veridical models for describing how faces invite real-world meaning.

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